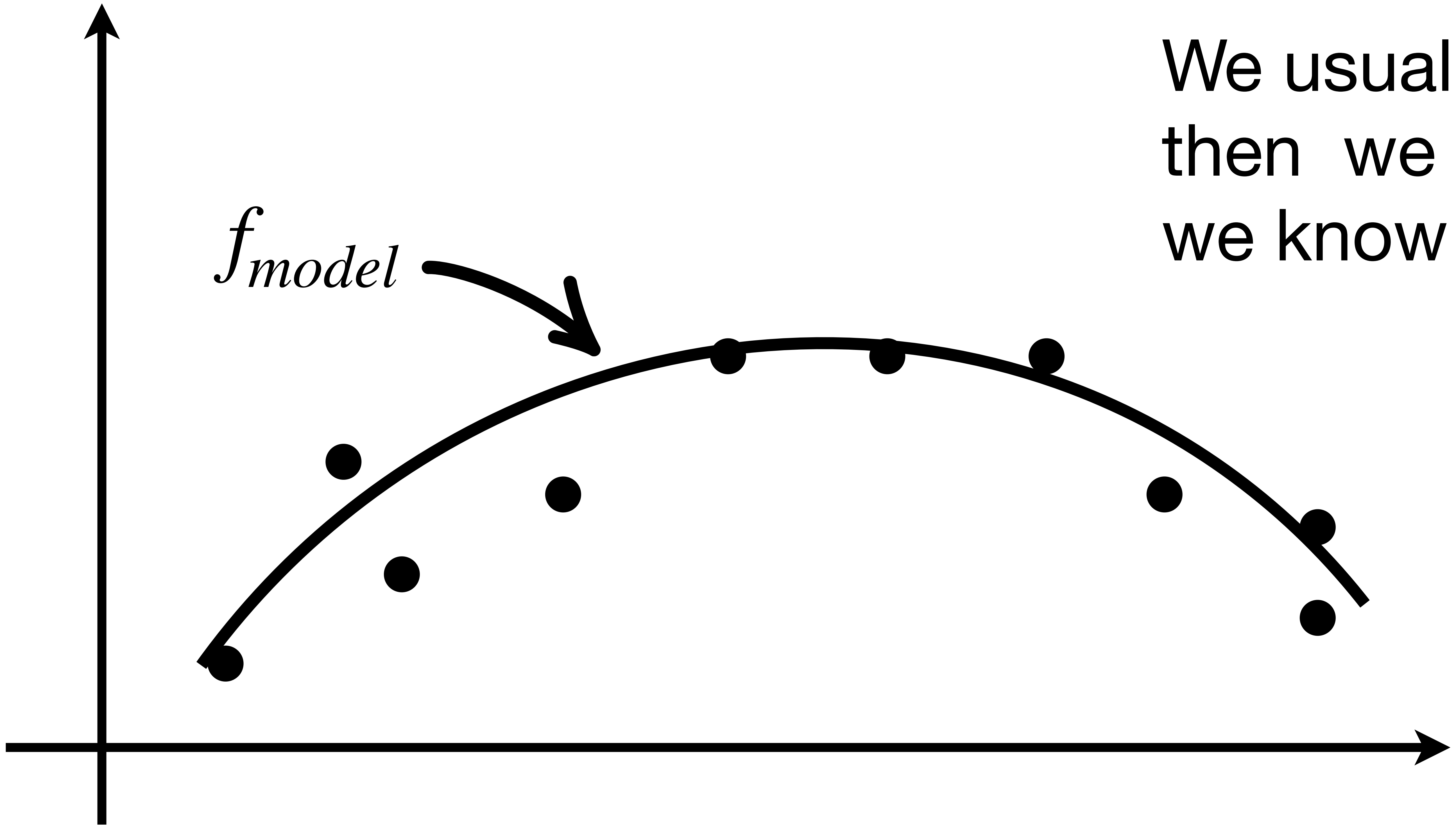
Implicit Problems (Models)

So far, we have looked at modeling data with explicit models.

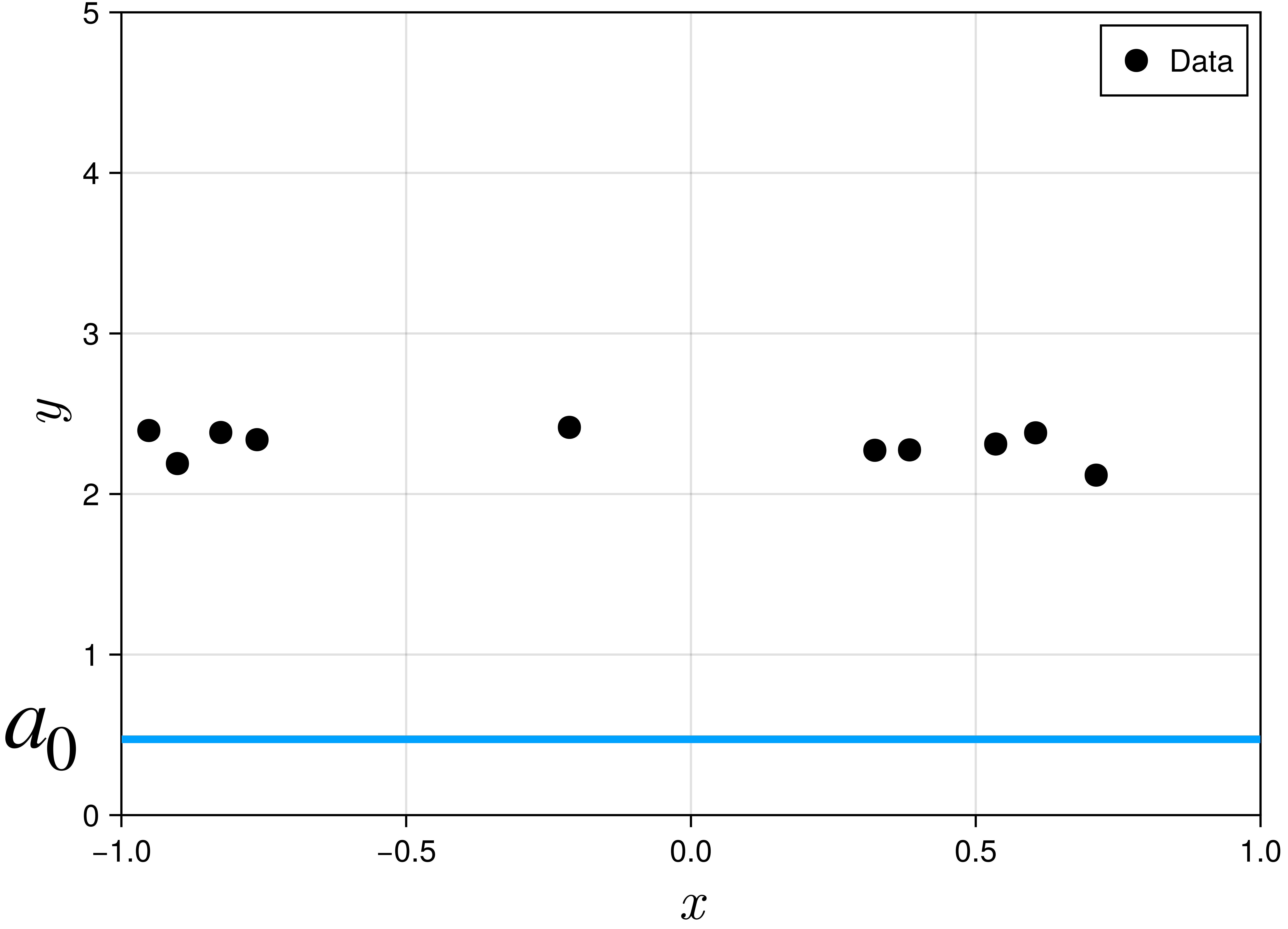
$$f_{model}(x, y) = a_0 + a_1 x + a_2 y + a_3 xy + a_4 x^2$$

$$f_{model}(x) = a_0 + a_1 x + a_2 x^2$$

$$f_{model}(x) = a_0$$



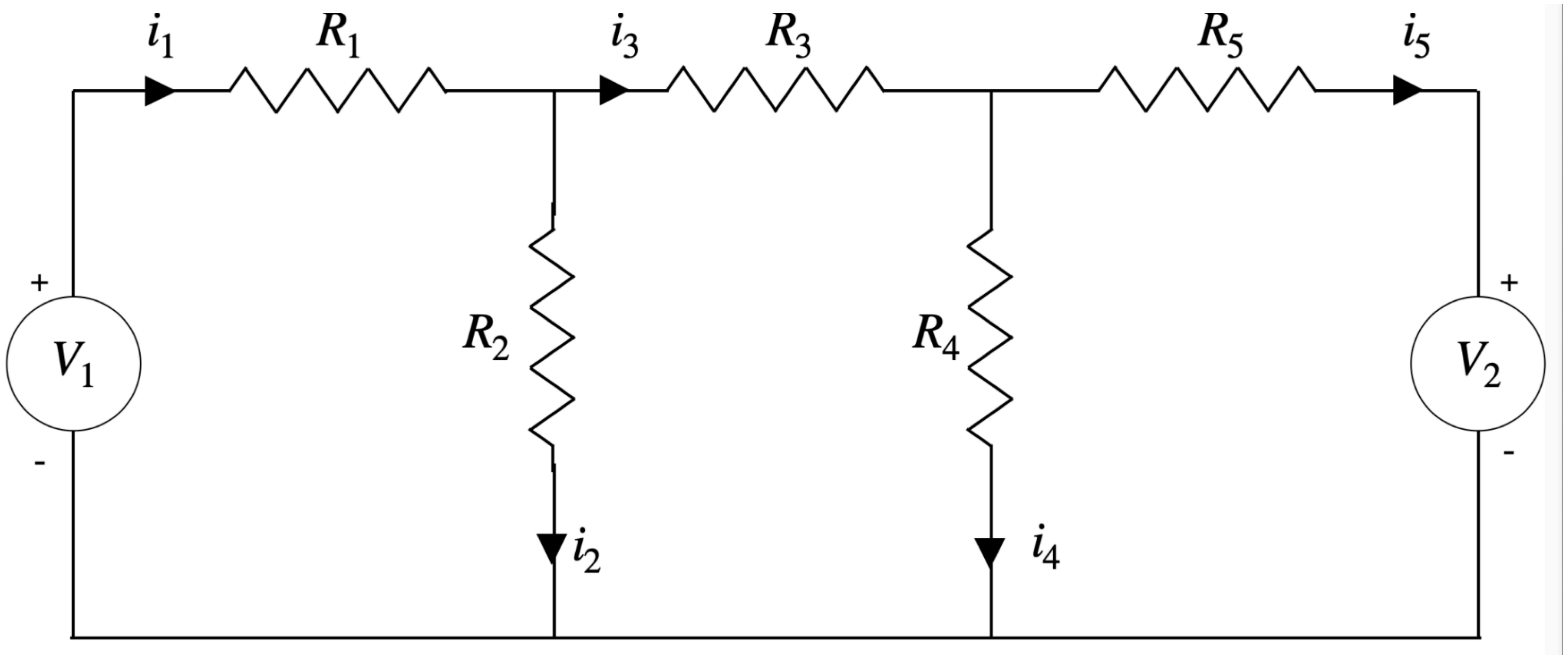
We usually (initially) guess all the $a_i's$ and then we can evaluate f_{model} (since we know all the x_i



$$f_{model} = a_0$$

$$S(a_0) = \frac{1}{2}\sum_{i=1}^{N}(y_i - a_0)^2$$

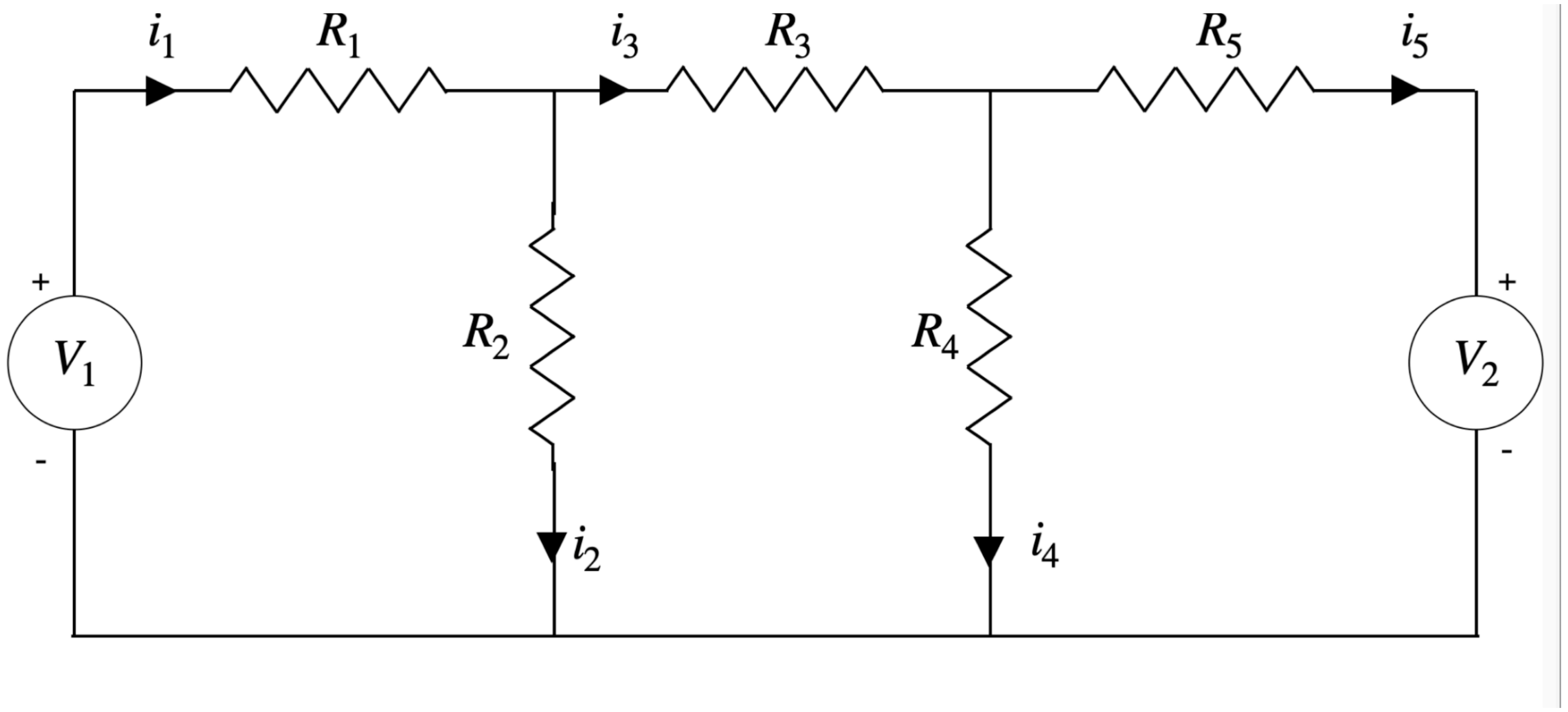
For many scientific applications, the output are defined implicitly e.g. for the below



Inputs: R_1, R_2, R_3, R_4, R_5 and V_1, V_2

Outputs: i_1, i_2, i_3, i_4 and i_5

we have to solve



Inputs: R_1, R_2, R_3, R_4, R_5 and V_1, V_2

Outputs: i_1, i_2, i_3, i_4 and i_5

We cannot explicitly write down the expressions for the outputs $i_1 \ldots i_5$. We have to solve

Outputs

Inputs

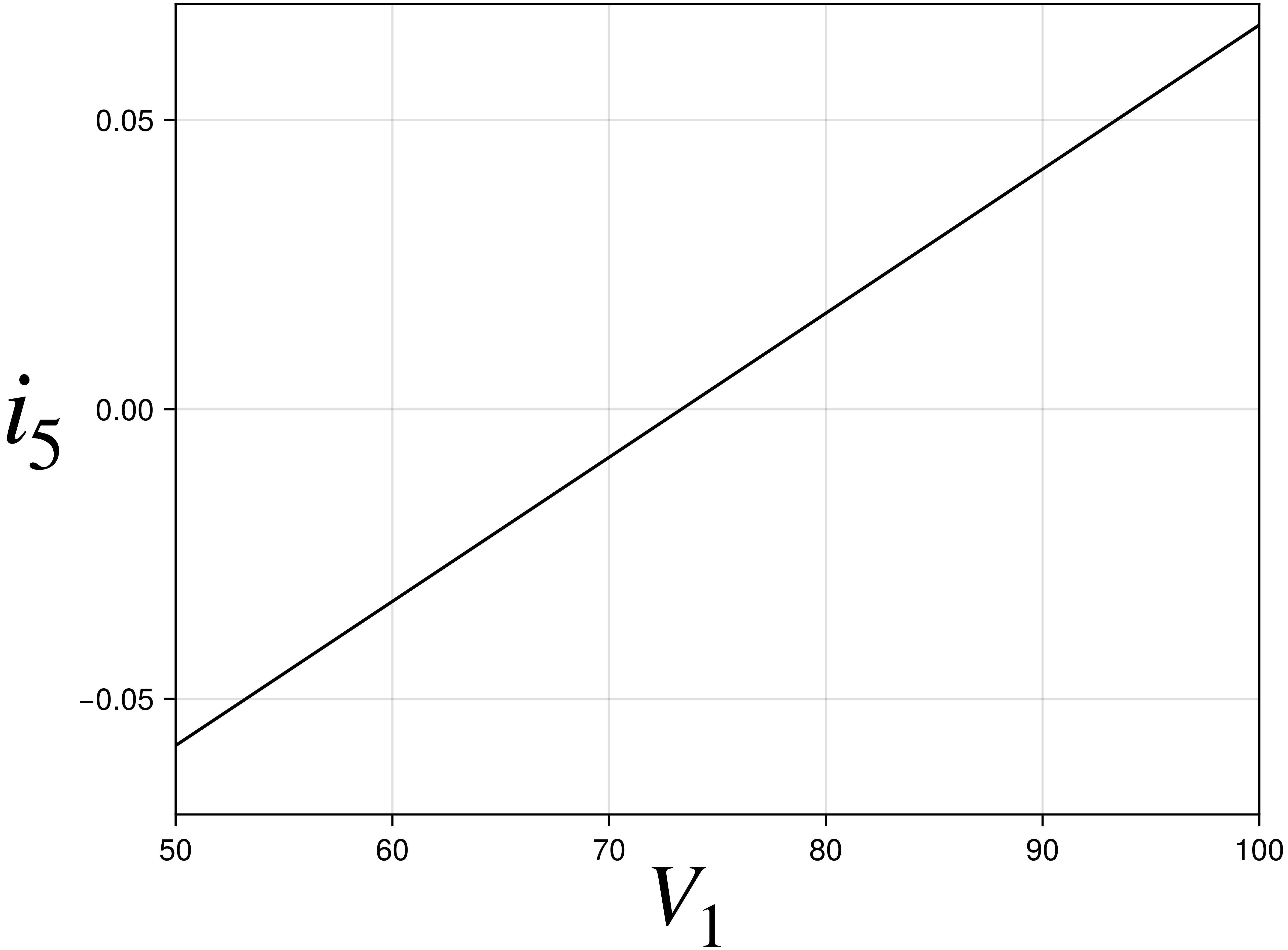
$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

Inputs

Suppose we know the value of i_5 $(=0)$ but we do not know V_1.

How do you find V_1?

We can run the solver for $V_1 \in [50, 100]$ to get



$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

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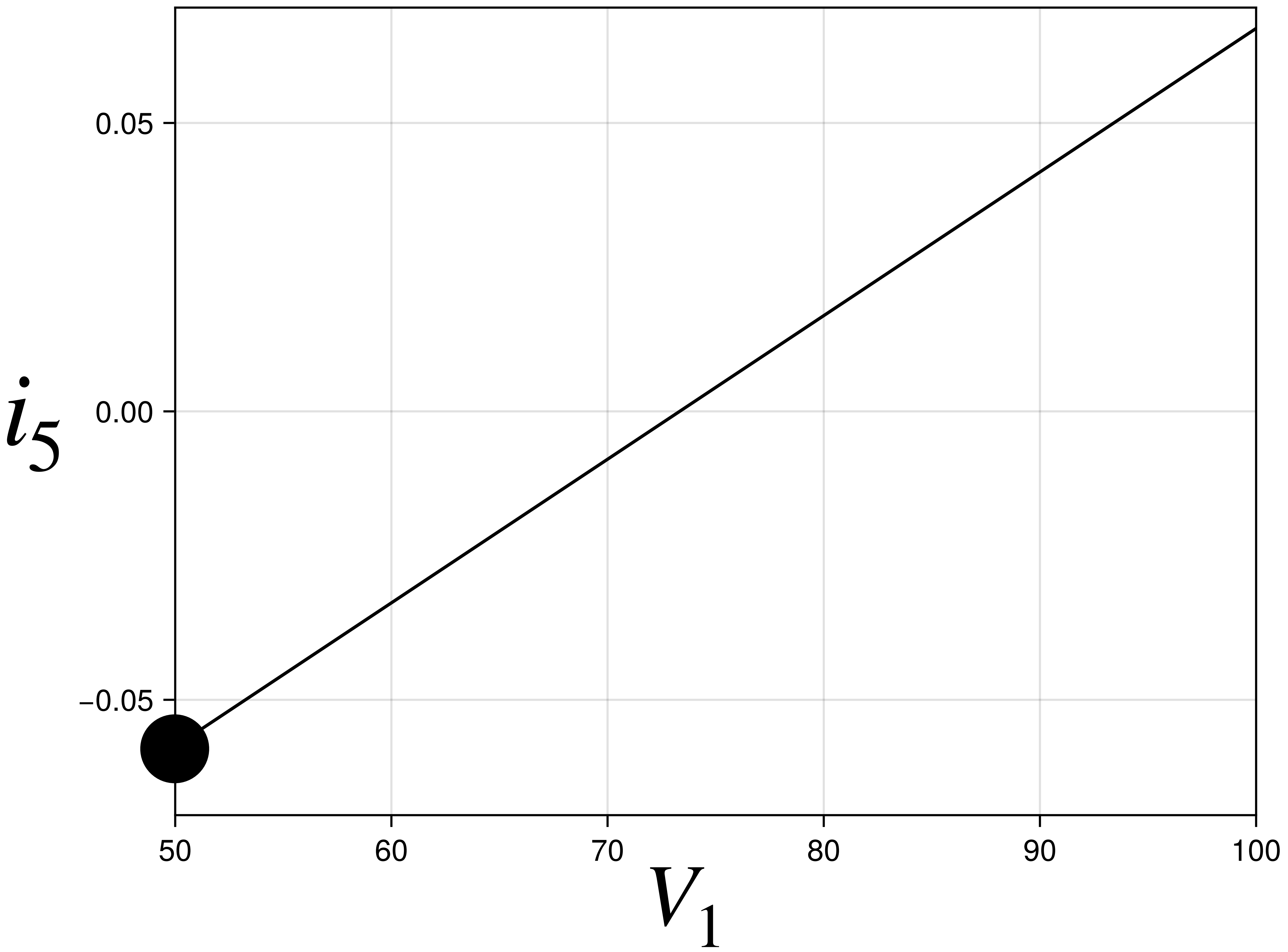

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

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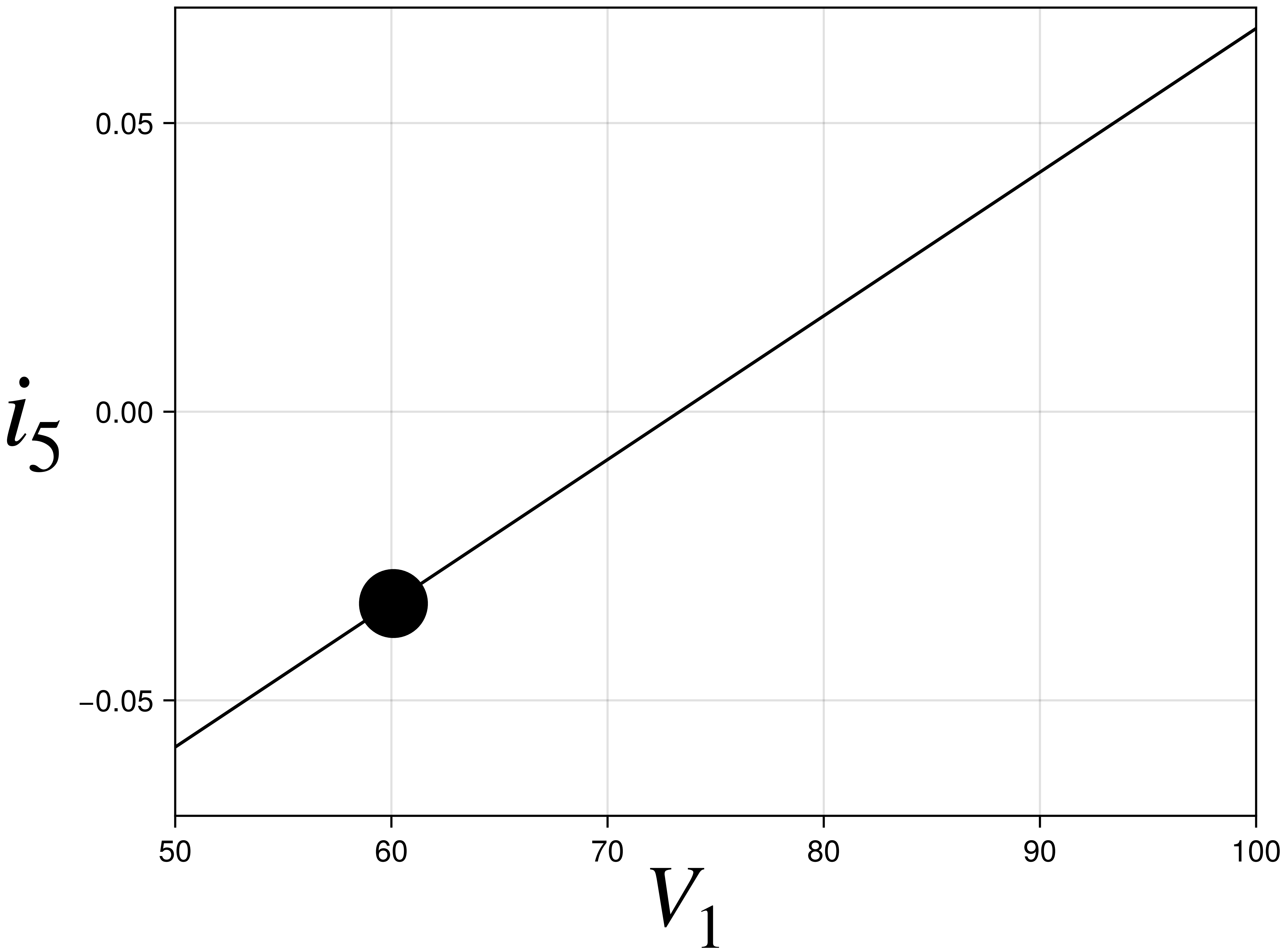

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} 60 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

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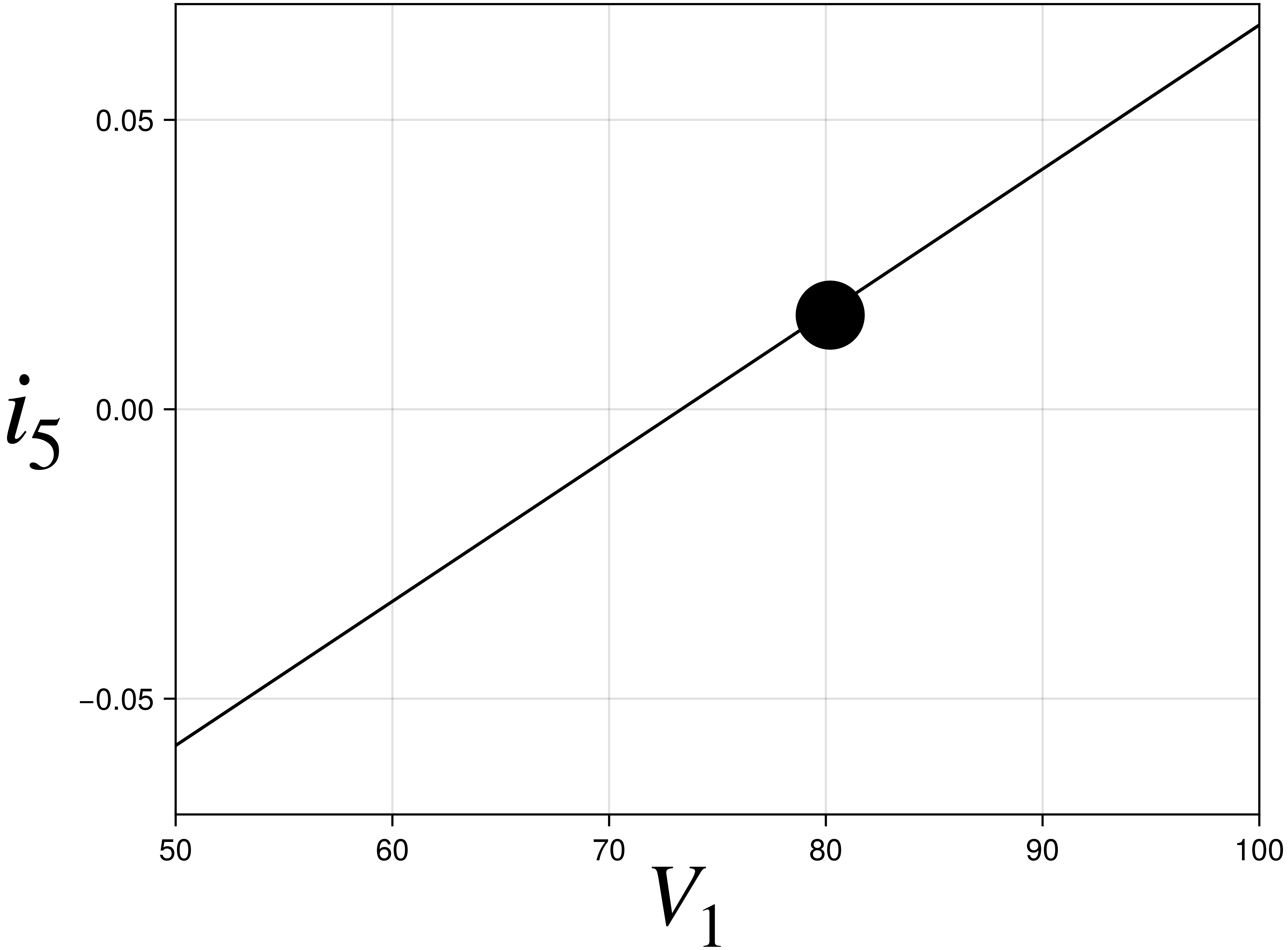

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} 80 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

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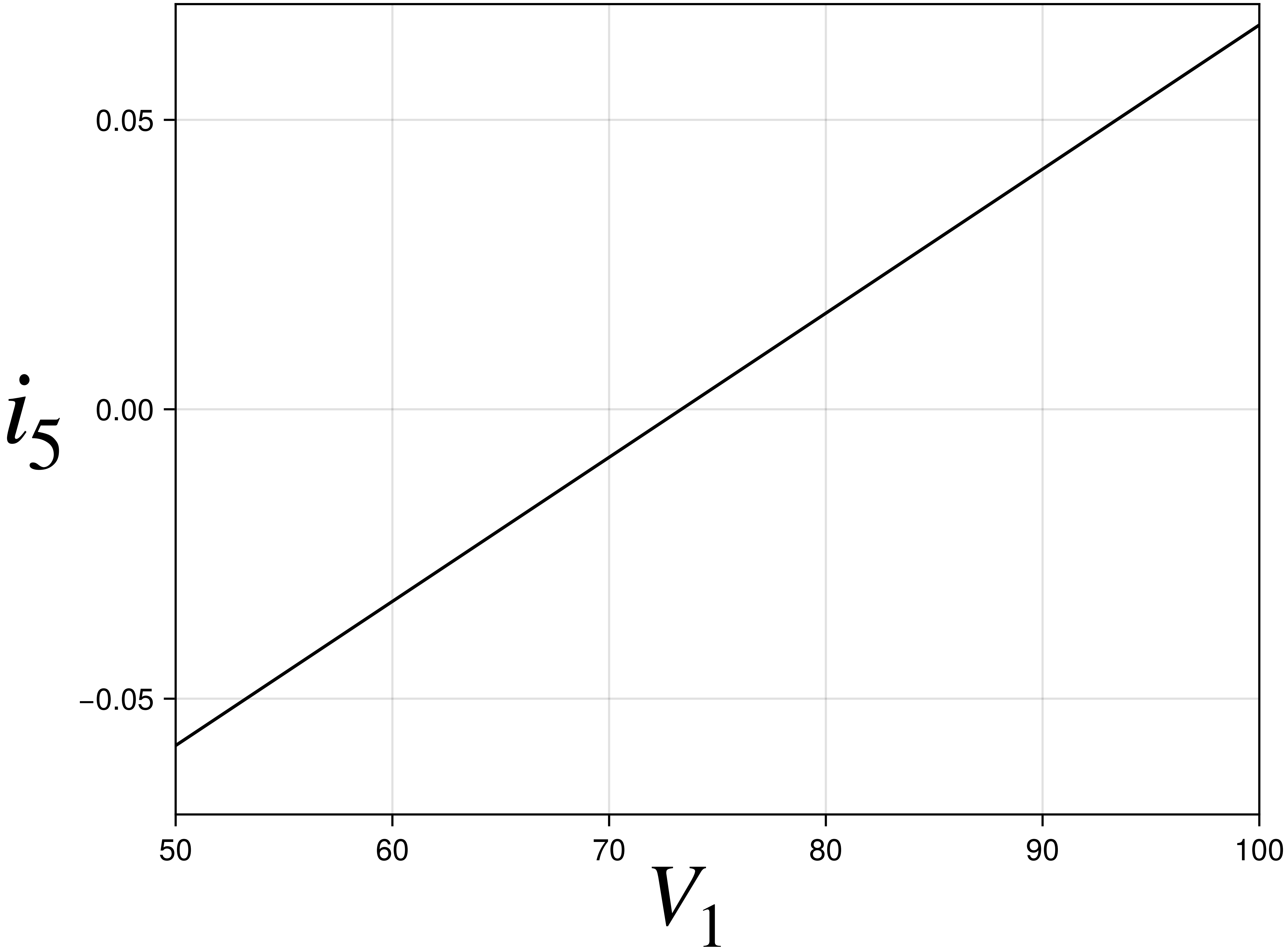
We can run the solver for $V_1 \in [50, 100]$ to get



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```
for i=1:length(V₁vec)
    C₁[1]=V₁vec[i]
    prob = LinearProblem(A,C₁)
    sol = solve(prob)
    i₅vec[i]=sol.u[5]
end
```

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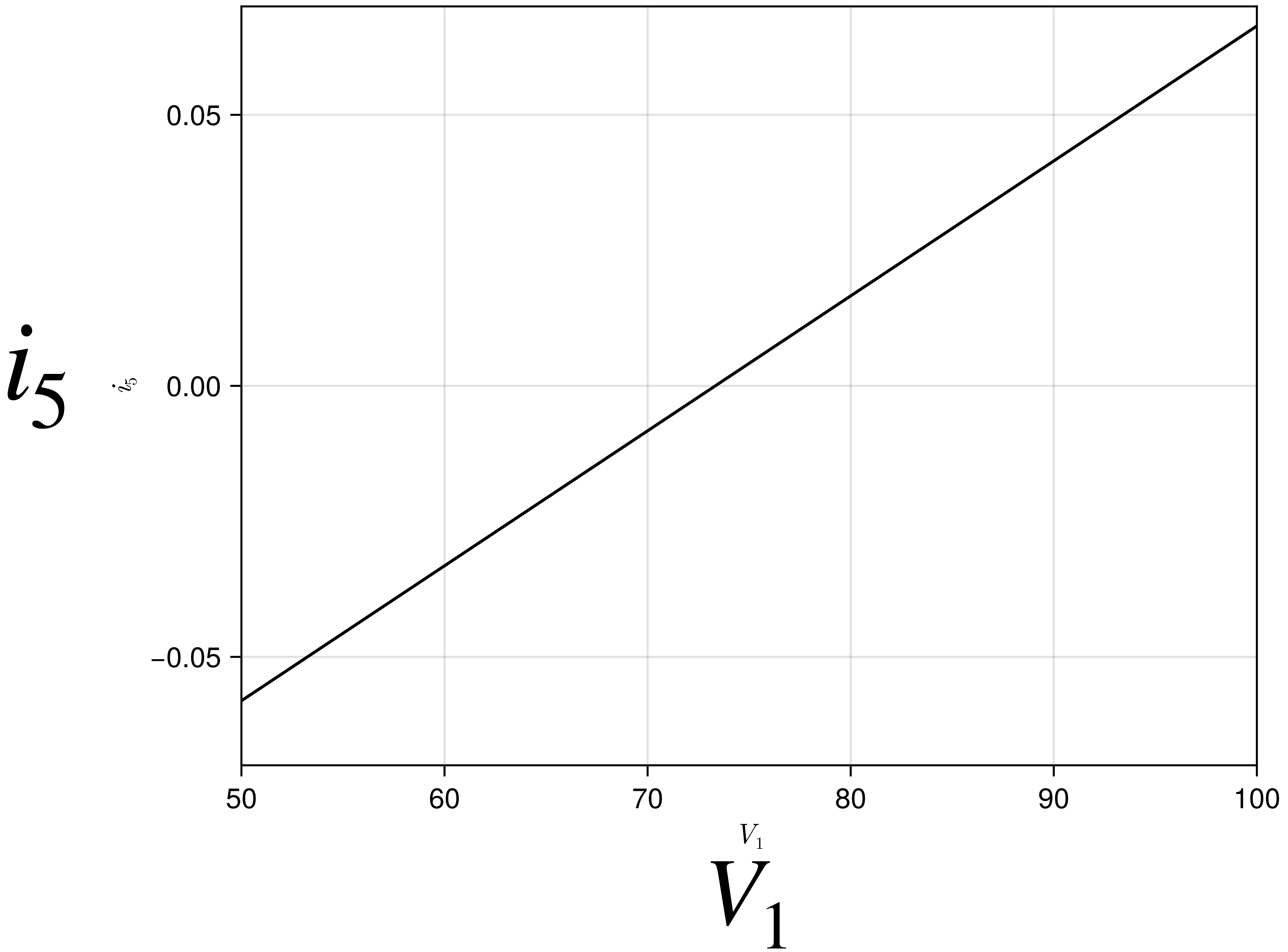

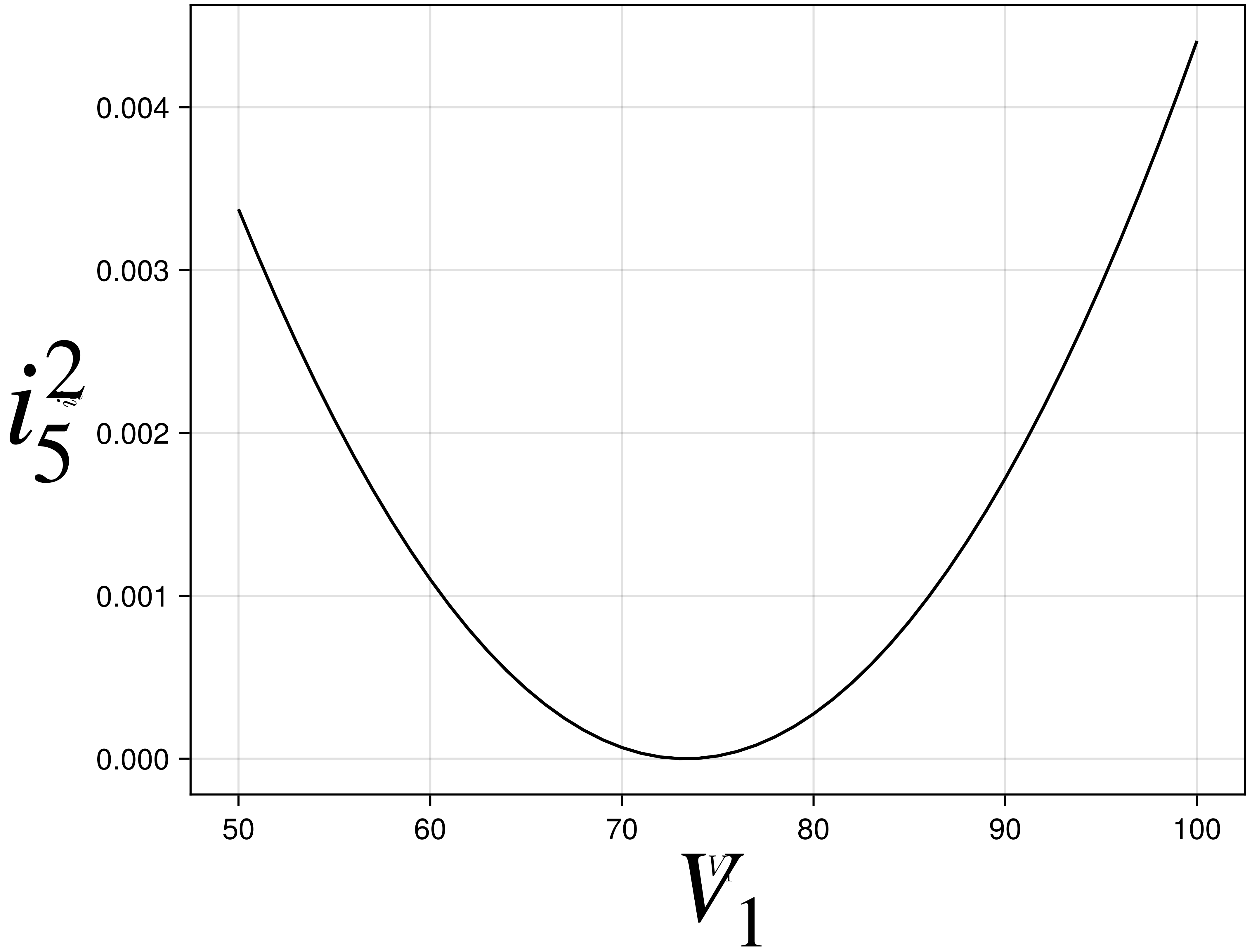

In the previous example(s) the predictions can be obtained easily if we know x and we guess a_0

$$\boxed{f_{model}(x)} = \boxed{a_0}$$

Now we have

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ \boxed{i_5} \end{Bmatrix} = \begin{Bmatrix} \boxed{V_1} \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

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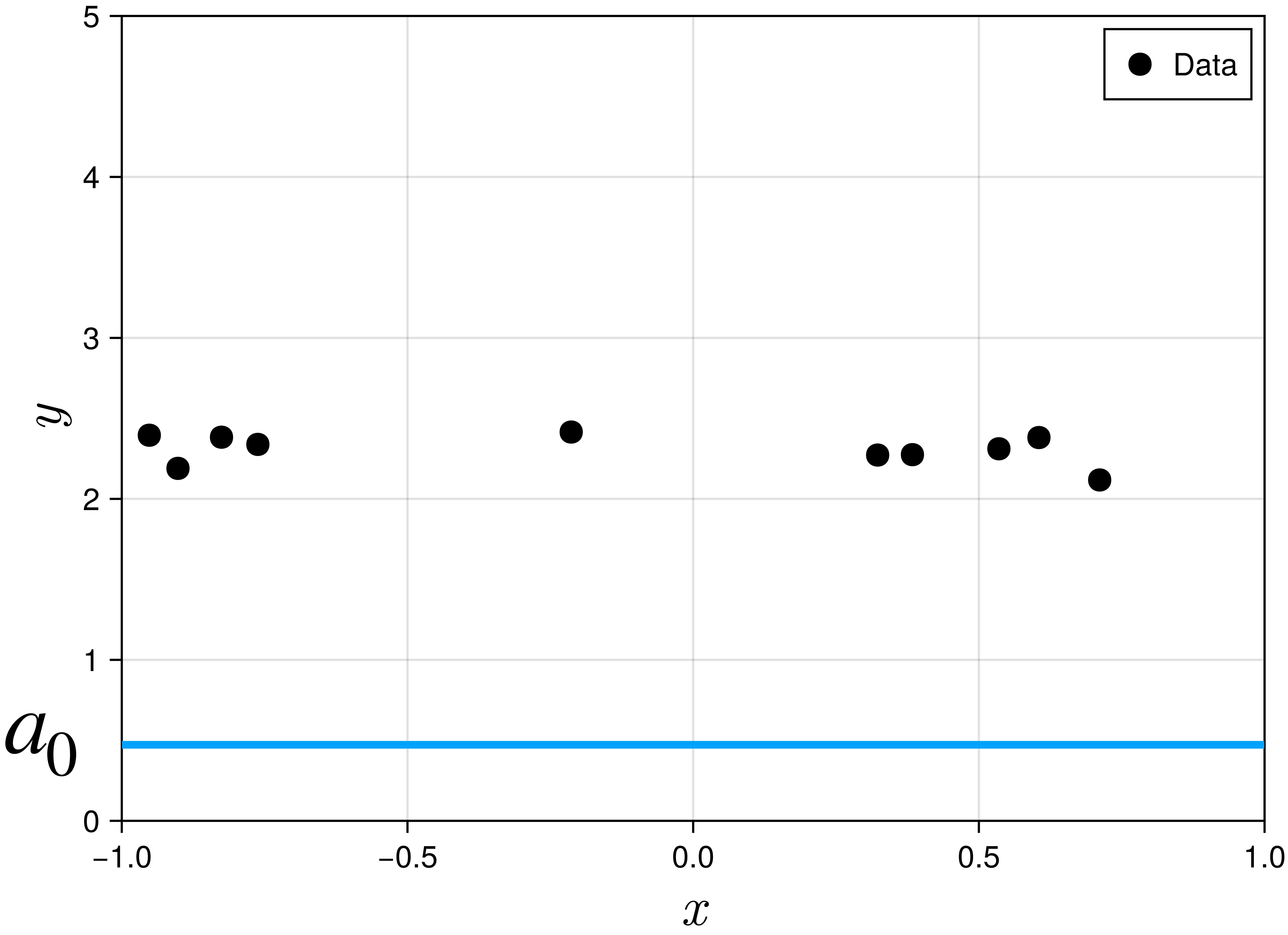
How do we calculate gradients??



$$f_{model} = a_0$$

$$S(a_0) = \frac{1}{2}\sum_{i=1}^{N}(y_i - a_0)^2$$

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$$\frac{\partial S}{\partial a_0} = -\sum_{i=1}^{N}(y_i - a_0)$$

Or we can use automatic differentiation (declare a_0 as a Dual number)

But now we have

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

$$S(V_1) = \frac{1}{2} \sum_{i=1}^{N} (0 - i_5(V_1))^2$$

i_5 is now an
Implicitly dependent on V_1!

How do we calculate $\dfrac{\partial S}{\partial V_1}$??

JUST USE AUTOMATIC DIFFERENTIATION

```julia
function S(p)
    C=[p[1],0,V₂,0,0]
    prob = LinearProblem(A,C)
    i = LinearSolve.solve(prob)
    return 0.5*sum(i[end].^2)
end
```

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

$$S(V_1) = \frac{1}{2}\sum_{i=1}^{N}(0 - i_5(V_1))^2$$

$$\frac{\partial S}{\partial V_1}$$

```julia
fmodel(a,x)=a[1];
```

$$f_{model} = a_0$$

```julia
function S(parameters,x_samples)
        ŷ = fmodel(parameters, x_samples)
        sum(0.5*(y_samples .- ŷ).^2)
end
```

$$S(a_0) = \frac{1}{2}\sum_{i=1}^{N}(y_i - a_0)^2$$

$$\frac{\partial S}{\partial a_0}$$

```julia
ForwardDiff.gradient(S,[parameters])[1]
```